

Building $D_{-\nu}$: the negative-order parabolic cylinder function in Lean

Claude, with Daniel Murfet

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Abstract

Unit 6 of the grammar-paper §4 autoformalisation, and the first step of the option-B plan to build the parabolic-cylinder infrastructure Mathlib lacks. We define the negative-order function $D_{-\nu}(x) = e^{-x^2/4}/\Gamma(\nu) \cdot \int_0^\infty u^{\nu-1} e^{-u^2/2-xu} du$ ($\nu > 0$) via its DLMF §12.5(i) integral representation — exactly the orders $D_{-2\lambda}$ that `cor:fluctuation_closed_form` needs — prove the DLMF integrand integrable on $(0, \infty)$, and pin the value $D_{-\nu}(0) = 2^{\nu/2-1}\Gamma(\nu/2)/\Gamma(\nu)$ as a sanity anchor. Zero `sorry/axiom`.

1 Setting

Units 1–5 formalised the whole of §4.1: Watanabe’s fluctuation function $S_\lambda(a)$, its integrability and base value, the derivative ladder, the IBP recurrence and ODE, the boundary case, and the lowering identity. What remains of §4 — the closed form $S_\lambda(a) = 2^{1-\lambda}\beta^{-\lambda}\Gamma(2\lambda)e^{\beta a^2/8}D_{-2\lambda}(-a\sqrt{\beta/2})$ and the QHO identification — rests on the parabolic cylinder function D_ν , which Mathlib does not have. So the arc forks: rather than defer, we build the piece of D_ν theory the paper actually uses. The negative-order integral representation is the right entry point — it is elementary (a Gamma-type integral, no Weber ODE or Hermite expansion needed) and it is precisely the branch the closed form lands in, since $2\lambda > 0$.

2 The things formalised

```
noncomputable def parCylNeg ( x : ) : :=
  Real.exp (-x ^ 2 / 4) / Real.Gamma *
  u in Set.Ioi (0 : ), u ^ ( - 1) * Real.exp (-u ^ 2 / 2 - x * u)

theorem parCylNeg_integrableOn ( x : ) (h : 0 < ) :
  IntegrableOn (fun u => u ^ ( - 1) * Real.exp (-u ^ 2 / 2 - x * u)) (Set.Ioi 0)

theorem parCylNeg_zero ( : ) (h : 0 < ) :
  parCylNeg 0 = 2 ^ ( / 2 - 1) * Real.Gamma ( / 2) / Real.Gamma
```

3 Proof strategy

Integrability. The integrand $u^{\nu-1}e^{-u^2/2-xu}$ is dominated by a pure Gaussian: the phase satisfies $-u^2/2 - xu \leq x^2 - u^2/4$, which is $(u/2 + x)^2 \geq 0$ rearranged. So the integrand is $\leq e^{x^2} \cdot u^{\nu-1}e^{-u^2/4}$, and the majorant is integrable by `integrableOn_rpow_mul_exp_neg_mul_rpow`

(the general Gamma-integral integrability at $p = 2$, $q = \nu - 1 > -1$, $b = \frac{1}{4}$). `Integrable.mono'` closes it, with the pointwise bound via `nlinarith [sq_nonneg (u/2+x)]` and the `rpow/npow` bridge $u^{(2:\mathbb{R})} = u^2$.

Value at 0. At $x = 0$ the integral is the standard $\int_0^\infty u^{\nu-1} e^{-u^2/2} du = 2^{\nu/2-1} \Gamma(\nu/2)$, an instance of `integral_rpow_mul_exp_neg_mul_rpow` at $p = 2$, $b = \frac{1}{2}$, giving $(\frac{1}{2})^{-\nu/2} \cdot \frac{1}{2} \cdot \Gamma(\nu/2)$; the `rpow` identity $(\frac{1}{2})^{-\nu/2} = 2^{\nu/2}$ and $2^{\nu/2-1} = 2^{\nu/2} \cdot \frac{1}{2}$ reconcile it to the stated form. The prefactor $e^{-0^2/4}/\Gamma(\nu) = 1/\Gamma(\nu)$ then gives the quotient.

4 Roadblocks and resolutions

All Lean bookkeeping, no mathematical friction:

- `integral_rpow_mul_exp_neg_mul_rpow` takes its hypotheses in the order $(0 < p)$, $(-1 < q)$, $(0 < b)$; feeding q 's bound where p 's was expected left a stray $\vdash 0 < \nu$. The `integrableOn` sibling happens to accept q 's bound first, so the two are *not* argument-compatible — check each signature rather than copying.
- The `rpow` arithmetic $(\frac{1}{2})^{-\nu/2} = 2^{\nu/2}$ needs `Real.inv_rpow` then `Real.rpow_neg+inv_inv`; `Real.rpow_neg_one` does not exist on this pin (a known repo gotcha), so 2^{-1} went through `Real.rpow_neg+Real.rpow_one`.
- `setIntegral_congr_fun` hands the pointwise goal as applied lambdas; `beta_reduce` before the `rw`, and rewrite the `exp`-argument directly (`ring` cannot pass under `exp`).
- The final `rw`-chain closes the intermediate integral value by `rfl`, so a trailing `ring` errored “no goals” — deleted; the outer prefactor step keeps its `ring`.

5 What was Mathlib, what was new

Both Gamma-integral engines (`integral_/integrableOn_rpow_mul_exp_neg_mul_rpow`) are Mathlib. The new content is the definition itself — Mathlib has no parabolic cylinder function — and the two elementary facts that make it usable downstream: an integrability certificate valid for *every* x (via the completed-square Gaussian majorant, so later change-of-variables arguments have their dominating function ready) and the closed value at the origin.

6 Lessons

When a paper needs a special function absent from the library, the negative-order / integral-representation branch is usually the cheapest foothold: it is a Gamma-type integral, so the existing $\int x^q e^{-bx^p}$ machinery does all the analytic work, and one completed square gives an x -uniform majorant that unlocks the substitution proofs to come. Build the representation the target formula actually consumes, not the “canonical” definition.

7 Follow-ups

Next unit: `cor:fluctuation_closed_form` itself — the change of variables $u = \sqrt{2\beta t}$ in S_λ , matching the kernel against `parCylNeg (2λ) (-a√β/2)` (scale by `integral_comp_mul_left_Ioi`,

then `integral_comp_rpow_Ioi_of_pos` backwards at $p = 2$, with Γ left outside the substitution). Then `eq:Dn_hermite` (D_n at integer order via Mathlib's Hermite polynomials) and `lem:qho`.